

AI Forecasting Demo: Global Oil Prices (2025–2030)

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Demo Overview

This AI-powered demo showcases a Transformer-based deep learning model trained on global oil market data from 2010 to 2024

- Crude Oil Prices (USD per barrel)
- Natural Gas Prices
- Global Oil Production (Million Barrels per Day)
- OECD Petroleum Stock Levels

Oil price data: Yahoo Finance

Other sources: U.S. Energy Information Administration (EIA)

Expo Use Case

This demo is part of Nayman AI & Data Consulting's showcase at the World AI Technology Expo

Visitors can explore a pre-trained forecast or upload a CSV to generate new predictions

The app offers interactive graphs, annotations, and downloadable summaries.

Why Use a Pre-Trained Transformer?

To ensure a fast and responsive experience for Expo visitors, the model is pre-trained using historical data

Training deep learning models like Transformers in real-time can take 1–2 minutes, which is not ideal during live demos

Pre-training ensures immediate access to high-quality results while still offering the flexibility to upload a custom dataset

This strikes the perfect balance between performance and interactivity.

Key Highlights

- Forecasting horizon: Jan 2025 to Dec 2030
- Model: Transformer (input: 36 months, output: 60 months)
- Built with Darts + Streamlit + Hugging Face
- Uses covariates + time features (year, month)
- Interactive + supports live CSV upload